Mood-Mate Robot 

CSC 317 1.5 – Human Computer Interaction

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Abstract

Human Computer Interaction (HCI) aims to develop systems that can understand and respond to human emotions to improve user experience and emotional well-being. This project presents Mood-Mate, an emotion-responsive interactive robot designed to enhance human mood through visual expression, physical interaction, and music-based feedback. The system detects human emotions using facial emotion recognition powered by the DeepFace framework and classifies them into four moods: Happy, Sad, Angry, and Neutral.

A Node.js server deployed on the Render cloud platform acts as the central communication hub, transmitting mood and positional data to multiple ESP32 microcontrollers. OLED displays are used to represent mood-based eye expressions, while a servo-controlled neck mechanism rotates toward the user based on angle data obtained from an external webcam. Additionally, a separate ESP32-based Bluetooth speaker system plays customized Spotify playlists corresponding to the detected mood. The robot operates using a direct power supply to ensure stable and continuous performance.

Experimental evaluation and real-world testing demonstrated reliable emotion detection, responsive physical movement, and synchronized music playback with minimal latency. User feedback indicated increased engagement and positive emotional response due to the combination of visual, physical, and auditory interaction. The results confirm that multi-modal emotion-aware systems can significantly enhance human robot interaction. This project highlights the potential application of emotion-responsive robots in mental well-being support, assistive technologies, and social robotics.

1. Introduction

1.1. Background and Motivation

Human Computer Interaction (HCI) focuses on creating systems that can understand and respond to human emotions to improve user experience. Emotional stress and mood imbalance are common in daily life, and music is widely recognized as an effective method for mood regulation. Motivated by this, our project aims to design an emotion-responsive interactive robot that detects human emotions and responds through visual expressions, physical interaction, and mood-based music playback. The goal is to create a more empathetic and engaging human–robot interaction system.

1.2. Problem Definition

Most existing robotic systems lack effective emotional responsiveness and personalized interaction. They often fail to adapt their behavior based on the user's emotional state, reducing their usefulness in emotional support scenarios. There is a need for a system that can detect human mood, respond visually and physically, and actively help improve the user's emotional state through interaction, rather than remaining passive.

1.3. Project Scope

This project involves the development of a powered (non-wireless) emotion-responsive robot system. The robot detects four human moods—Happy, Sad, Angry, and Neutral—and responds accordingly. A Node.js server, deployed on Render, processes mood data and sends it to ESP32 microcontrollers. OLED displays are used to change the robot's eye expressions based on the detected mood. An external webcam is used to detect human position, with angle data sent through the Node.js server to rotate the robot's neck toward the user. Additionally, a separate ESP32-based Bluetooth speaker system plays customized Spotify playlists corresponding to each detected mood.

1.4. Significance of the Project

The project demonstrates how emotion-aware systems can enhance human robot interaction by combining emotion detection, visual feedback, physical movement, and music-based mood regulation. By integrating Spotify-based mood playlists and interactive behaviors, the robot actively attempts to improve the user's emotional state rather than simply displaying emotions. This work contributes to HCI research by showcasing a practical, multi-modal emotional interaction system with potential applications in mental well-being, assistive technologies, and social robotics.

2. Objectives

2.1. Main Objectives

- To design and develop an emotion-responsive interactive robot that enhances human mood through music-based interaction.
- To apply Human Computer Interaction principles to create a robot that can perceive human emotions and respond using visual, physical, and auditory feedback.

2.2. Specific Objectives

- To detect and classify human emotional states into four moods: Happy, Sad, Angry, and Neutral.
- To develop a Node.js server (deployed on Render) to process mood and angle data and communicate with ESP32 microcontrollers.
- To implement OLED eye animations that dynamically change according to the detected mood.
- To use an external webcam to detect human position and calculate angles for rotating the robot's neck toward the user.
- To develop a Bluetooth speaker system using a separate ESP32 to play customized Spotify playlists based on the detected mood.

- To integrate all components into a stable, direct powered (non-wireless) robotic system for consistent operation.

2.3. Expected Deliverables

- A fully functional emotion-responsive robot capable of interacting with users in real time.
- A Node.js server deployed on Render that successfully manages mood and angle data communication.
- OLED eye display system showing mood-based expressions.
- A neck-rotation mechanism that tracks and faces the user using external webcam data.
- A Bluetooth speaker system that plays mood-specific Spotify playlists.
- Complete project documentation including system architecture, implementation details, and evaluation results.

3. Tools and Technologies

3.1. Hardware Components

- **ESP32 Microcontrollers** – Used for controlling OLED displays, servo motor for neck rotation, and Bluetooth speaker functionality.
- **OLED Display** – Used to display dynamic eye expressions based on detected human mood.
- **External Webcam** – Used to detect human position and calculate angles for rotating the robot's neck toward the user.
- **Servo Motor** – Controls the physical rotation of the robot's neck to face the user.
- **Bluetooth Speaker Module** – Implemented using a separate ESP32 to play mood-based audio output.
- **Direct Power Supply** – Provides stable power to the robot (non-wireless operation).

3.2. Software Platforms

- **Node.js Server** – Handles mood and angle data communication between the detection system and ESP32 devices.
- **Render Cloud Platform** – Used to deploy the Node.js server for real-time data exchange.
- **Spotify Platform** – Used to stream and play customized playlists based on detected moods.
- **PlatformIO** – Used for programming ESP32 microcontrollers.

3.3. Development Tools

- **Visual Studio Code (VS Code)** – Primary code editor for Node.js, Python, and ESP32 development.
- **Python Environment (Virtual Environment)** – Used for running mood detection and webcam processing applications.
- **Git / GitHub** – Used for version control and project management.
- **Serial Monitor** – Used for debugging ESP32 communication and sensor data.

3.4. Libraries & Frameworks

- **DeepFace** – Used for facial emotion recognition and mood detection.
- **OpenCV** – Used for real-time webcam video processing and face tracking.
- **Express.js** – Backend framework used in the Node.js server.
- **ESP32 Arduino Core** – Provides hardware abstraction and communication support for ESP32.
- **Servo Library** – Used to control servo motor movement for neck rotation.
- **Bluetooth Libraries (ESP32)** – Used to implement Bluetooth audio output functionality.

4. Methodology

4.1. System Architecture Overview

The system follows a multi-layered architecture integrating emotion detection, physical interaction, and music playback. A Node.js server, deployed on Render, acts as the central hub for receiving mood data from the laptop webcam, calculating human angles from the external webcam, and sending this information to two ESP32 microcontrollers. One ESP32 controls the OLED eye expressions and servo motor for neck rotation, while the other drives a Bluetooth speaker to play mood-based Spotify playlists.

4.2. Hardware Setup

The hardware setup of the system consists of multiple components working together to enable real-time emotional interaction. Two ESP32 microcontrollers are programmed separately, where one ESP32 is responsible for controlling the OLED eyes and the servo motor, while the other ESP32 manages the Bluetooth speaker functionality. The OLED eyes are used to display dynamic eye expressions that change according to the detected human mood. A servo motor, combined with an external webcam, is used to track the human position and rotate the robot's neck toward the user in real time based on calculated angle data. Additionally, a Bluetooth speaker plays customized Spotify playlists corresponding to the four detected moods: Happy, Sad, Angry, and Neutral. All hardware components are powered using a direct power supply to ensure stable and continuous operation of the system.

4.3. Data Collection Process

Facial images are captured in real time using the laptop webcam for emotion detection, while the external webcam captures the user's position relative to the robot to calculate angles required for neck rotation. Both the emotion data and positional information are sent to the Node.js server, which processes this data and forwards it for subsequent actions such as updating eye expressions, rotating the robot's neck, and triggering mood-based music playback.

4.4. Machine Learning Model Development

DeepFace is used for facial emotion recognition in the system. The model classifies detected facial expressions into four emotion categories: Happy, Sad, Angry, and Neutral. Real-time video frames captured from the webcam are pre-processed and continuously fed into the model to perform ongoing mood prediction.

4.5. Model Deployment

The emotion detection system is integrated with the Node.js server, which processes incoming data and communicates with the ESP32 boards via Wi-Fi. Based on the real-time mood predictions, the system triggers corresponding OLED eye animations and selects the appropriate Spotify playlists for playback.

4.6. System Integration

The Node.js server handles all communication between the hardware and software modules in the system. The ESP32 responsible for the servo motor receives angle data from the server to rotate the robot's neck toward the user, while the second ESP32 plays the appropriate Spotify playlist through the Bluetooth speaker. As a result, the system operates as a fully integrated unit, responding to the user's mood and position through coordinated visual, auditory, and physical feedback.

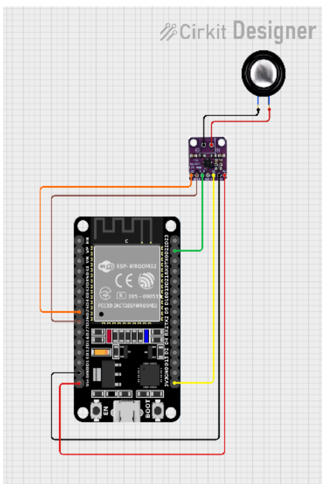
5. System Implementation

5.1. Hardware Implementation

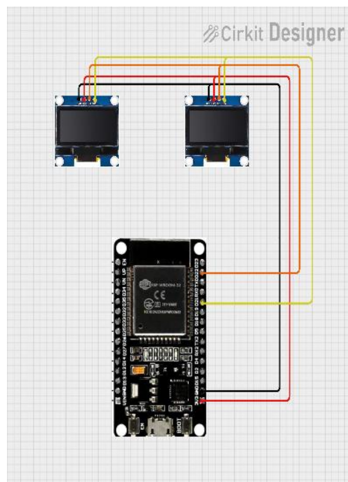
The hardware components of the robot were assembled to enable real-time emotional interaction and mood-based response:

- **ESP32 Microcontrollers:** Three ESP32 boards were programmed separately.
    - **ESP32 #1:** Controls the OLED eyes. It receives mood data data from the Node.js server to display the corresponding eye expressions.
    - **ESP32 #2:** Controls the Servo motor. It receives angle data from Node.js server to rotate the robot's neck toward the user.
    - **ESP32 #3:** Handles the Bluetooth speaker module, playing customized Spotify playlists for each mood.

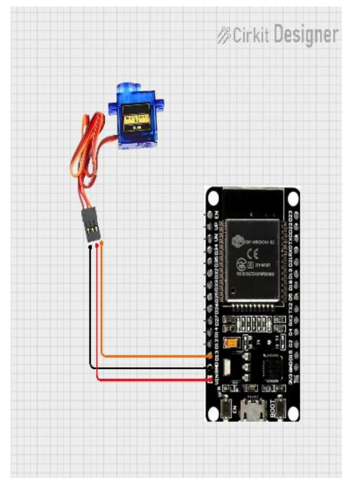
- **OLED Display:** Connected to ESP32 #1, it dynamically changes eye patterns based on detected moods (Happy, Sad, Angry, Neutral).
- **Servo Motor & External Webcam:** The servo motor, controlled by ESP32 #2, rotates the robot's neck based on angles calculated from the external webcam.
- **Bluetooth Speaker:** Connected to ESP32 #3 to stream mood-specific music via Spotify.
- **Direct Power Supply:** Ensures uninterrupted operation of all electronic components, as the robot is powered directly (non-wireless).



Bluetooth Speaker



OLED Display



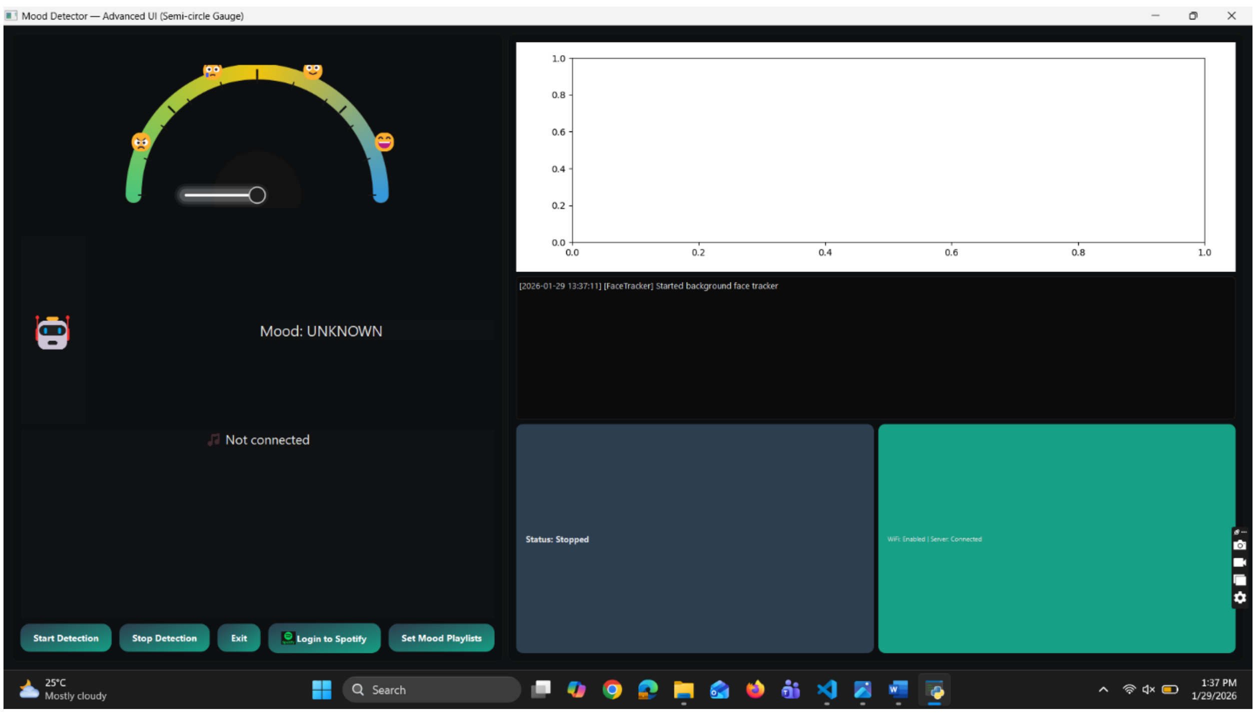
Servo Motor

5.2. Software Implementation

The software components were developed to process emotions, communicate with hardware, and provide real-time interactive responses:

- **Node.js Server:** Acts as the central hub deployed on Render. It receives mood predictions from the laptop webcam and user angle data from the external webcam. The server sends corresponding commands to both ESP32 boards.
- **Emotion Detection (DeepFace):** Facial images captured from the laptop webcam are processed using DeepFace to classify emotions into four categories.

- **Angle Detection (OpenCV):** The external webcam detects the user's position relative to the robot, calculates angles, and sends them to the Node.js server for servo motor control.
- **OLED Eye Display:** The Node.js server sends mood signals to ESP32 #1, which updates the OLED display with appropriate eye expressions.
- **Spotify Integration:** Based on the detected mood, the Node.js server communicates with ESP32 #2 to play mood-specific playlists through the Bluetooth speaker.



6. Testing & Evaluation

6.1. Performance Evaluation

The system performance was evaluated based on emotion detection accuracy, response time, and interaction consistency. The DeepFace-based emotion detection module was tested under varying lighting conditions and facial orientations to assess classification reliability across the four moods: Happy, Sad, Angry, and Neutral. System response time was measured from the moment an emotion was detected to the execution of actions, including OLED eye expression changes, neck rotation, and music playback. Servo motor movement accuracy was evaluated by comparing the angle values calculated from the external webcam with the actual physical neck rotation. Additionally, Bluetooth audio performance was tested to ensure stable playback and proper synchronization with detected mood changes.

6.2. Real-World Testing

The robot was tested in real-world interaction scenarios with users standing at varying distances and positions. During testing, users interacted naturally with the robot while their facial expressions and movements were detected in real time. The external webcam successfully tracked user position, allowing the robot to rotate its neck toward the user accurately. Mood-based Spotify playlists were triggered correctly according to the detected emotions, contributing to a more engaging and interactive experience. User feedback indicated improved engagement due to the combined effect of visual expressions, physical movement, and personalized music playback.

6.3. Error Handling & Debugging

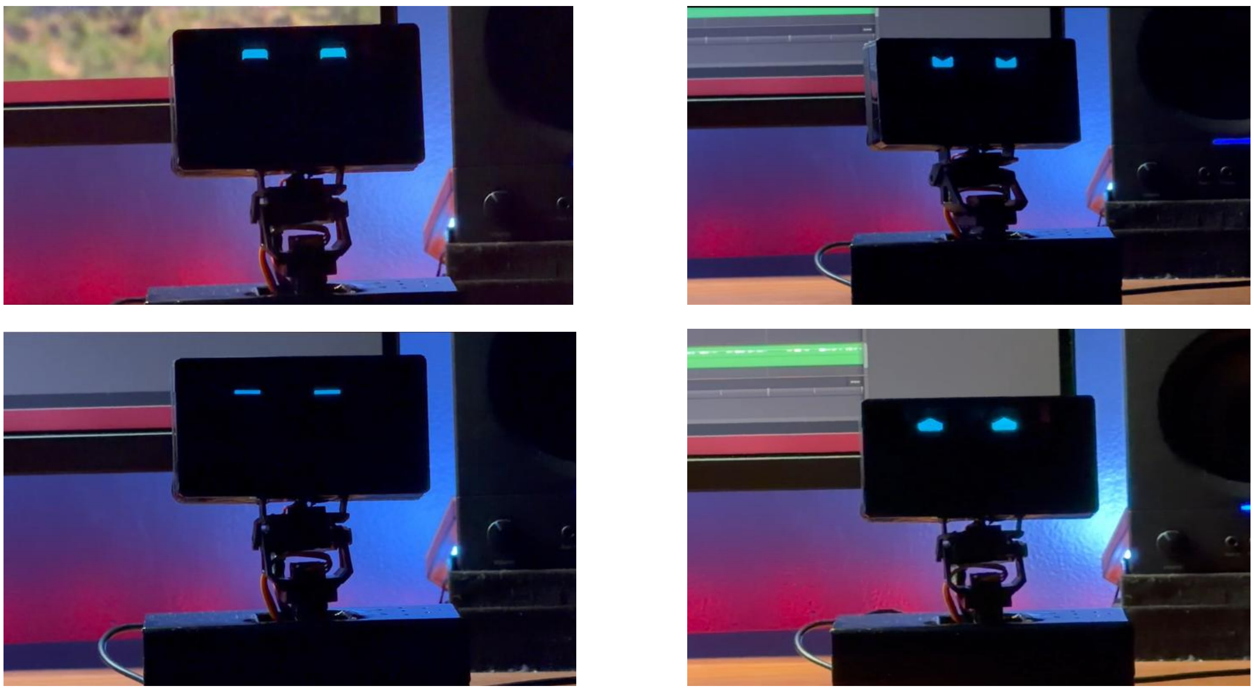
Robust error handling and debugging mechanisms were implemented to ensure overall system stability. Network interruptions between the Node.js server and ESP32 devices were managed using reconnection logic to maintain continuous communication. Invalid or uncertain emotion predictions were handled by assigning a default Neutral mood to prevent incorrect system responses. Servo motor angle limits were enforced to avoid mechanical damage during operation. Additionally, serial monitoring and logging were used throughout development and testing to identify and resolve communication, hardware, and timing-related issues.

7. Results

7.1. System Performance Summary

The developed system successfully demonstrated real-time emotion-responsive interaction. The DeepFace-based emotion detection module accurately classified facial expressions into four moods—Happy, Sad, Angry, and Neutral—under normal lighting conditions. The Node.js server efficiently transmitted mood and angle data to the ESP32 microcontrollers with minimal latency.

The OLED eye expressions updated smoothly according to mood changes, and the servo-controlled neck rotation accurately followed user position detected by the external webcam. The Bluetooth speaker system reliably played mood-specific Spotify playlists without noticeable interruptions. Overall, the system operated stably under continuous use, confirming effective integration of emotion detection, physical interaction, and music playback.



7.2. User Feedback & Observations

User interactions indicated high engagement and positive emotional response. Users reported that the combination of facial expressions, physical movement, and personalized music made the robot feel more interactive and emotionally aware. The neck rotation feature improved the sense of attention and presence, while the OLED eyes enhanced emotional expressiveness.

Many users observed that mood-based music helped improve or stabilize their emotional state, supporting the project's primary goal of mood enhancement through interaction. Some limitations were noted, such as reduced emotion detection accuracy under poor lighting or rapid facial movements, suggesting areas for future improvement.

8. Discussion

The results of this project demonstrate that integrating emotion recognition with physical and auditory responses can significantly enhance human–robot interaction. The system effectively combined facial emotion detection, visual expression through OLED eyes, physical movement via neck rotation, and mood-based music playback to create a more engaging and empathetic robotic experience.

The use of DeepFace enabled reliable real-time emotion classification for the four targeted moods. While performance was strong under normal conditions, variations in lighting and rapid facial movements affected accuracy, highlighting common challenges in vision-based emotion recognition. Despite these limitations, the system's fallback to a neutral mood ensured stable interaction and prevented abrupt or incorrect responses.

The physical interaction features, particularly the robot's neck rotation toward the user, increased the perception of attentiveness and presence. This aligns with HCI principles that emphasize responsiveness and embodiment in interactive systems. OLED eye animations further enhanced emotional communication, making the robot's reactions more intuitive and relatable.

Music playback through customized Spotify playlists played a central role in mood regulation. User feedback indicated that the music selection often helped improve or stabilize emotional states, supporting the project's primary objective of mood enhancement through interaction. The use of a separate ESP32 for Bluetooth audio improved modularity and reduced system complexity.

Overall, the project successfully demonstrates how multi-modal interaction combining visual, physical, and auditory feedback can improve emotional engagement in human robot systems. Future improvements could include more robust emotion classification, adaptive playlist selection based on user preferences, and additional interaction modalities such as voice or gesture recognition.

9. Future Improvements

9.1. Model Enhancements

Future work can improve emotion detection accuracy by training or fine-tuning models on more diverse datasets, including variations in lighting, facial orientation, and ethnicity. Incorporating temporal emotion analysis across multiple frames could reduce misclassification caused by rapid facial movements. Additionally, combining facial emotion recognition with other modalities such as voice tone or gesture recognition could provide more robust and reliable mood detection.

9.2. Hardware Optimization

The hardware design can be optimized by improving power management and reducing component size for a more compact system. Upgrading to higher-resolution cameras and smoother servo motors could enhance tracking accuracy and motion fluidity. Improved mechanical design for the neck rotation mechanism could further increase durability and responsiveness.

9.3. Feature Expansion Opportunities

The system can be expanded by introducing voice interaction, allowing the robot to communicate verbally with users. Personalized music recommendations based on user history and preferences could enhance mood regulation effectiveness. Additional emotional states beyond the current four moods could be incorporated, along with richer OLED animations and expressive movements to increase emotional realism.

9.4. Scalability Considerations

To support scalability, the system architecture can be extended to handle multiple robots communicating with a centralized server. Cloud-based processing could enable real-time updates and remote monitoring. Modular hardware and software design would allow easy upgrades and customization, making the system adaptable for larger deployments in environments such as therapy centers, educational institutions, or public spaces.

10. Conclusion

10.1. Summary of Work

This project successfully designed and implemented an emotion-responsive interactive robot that enhances human mood through music-based interaction. The system integrated facial emotion detection using DeepFace, physical interaction through neck rotation, visual expression using OLED eye displays, and auditory feedback via mood-based Spotify playlists. A Node.js server deployed on Render coordinated communication between software and hardware components, enabling real-time interaction.

10.2. Achievement of Objectives

All primary and specific objectives of the project were achieved. The robot accurately classified human emotions into four moods—Happy, Sad, Angry, and Neutral—and responded appropriately through synchronized visual, physical, and auditory feedback. The system demonstrated stable performance, effective integration of multiple ESP32 microcontrollers, and reliable real-time communication, confirming the feasibility of emotion-aware human robot interaction.

10.3. Impact and Potential Applications

The project highlights the potential of multi-modal interaction systems in improving emotional engagement and user experience. By actively responding to human emotions and attempting to regulate mood through music, the robot demonstrates potential applications in mental well-being support, assistive technologies, therapy environments, and social robotics. This work contributes to the field of Human Computer Interaction by showcasing how emotion-aware systems can create more empathetic and meaningful interactions between humans and machines.

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12. Contribution

Task / Component	AS2022431	AS2022432
Idea Generation & System Design	✓	✓
Face Detection & Emotion Analysis	✓	✓
Head Tracking & Servo Control	✓	✓
Backend Server (Node.js) Integration	✓	✓
Spotify API Integration	✓	✓
GUI Design	✓	✓
Documentation & Report Writing	✓	✓